

DATA CLEANING & DATA QUALITY VALIDATION REPORT

Employee Attendance Dataset — Data Quality Assessment

Objective: To identify, clean, standardize, and validate common data-quality problems in an employee attendance dataset.

Dataset description: A synthetic workforce attendance dataset containing attendance IDs, dates, employee IDs, employee names, departments, attendance status, working hours, and locations.

1. Original Dataset — Sample

Attendance ID	Date	Employee	Name	Dept.	Status	Hours	Location
ATT001	2026-02-01	E101	Arun	IT	Present	8	Chennai
ATT002	01/02/2026	E102	Meena	it	Present	8	Chennai
ATT003	2026-02-02	E103	Ravi	Finance	Present	7.5	Bengaluru
ATT004	2026-02-03		Divya	HR	Present	8	Hyderabad
ATT005	2026-02-03	E105	Kiran	IT	Present	-2	Chennai
ATT006	2026-02-04	E106	Sanjay	Finance	Absent	-8	Bengaluru
ATT007	2026-02-05	E107	Priya	HR	Present		Hyderabad
ATT008	2026-02-06	E108	Arun Kumar	IT	Present	8	Chennai
ATT009	2026-02-06	E108	Arun Kumar	IT	Present	8	Chennai
ATT010	06-02-2026	E109	Nisha	it	Present	9	Hyderabad

The sample illustrates the major quality problems found in the full synthetic dataset of 30 attendance records.

2. Data Quality Issues Identified

Issue	Example	Recommended treatment
Missing employee ID	ATT004 has blank Employee ID	Flag for review / retain as Unknown
Duplicate attendance	ATT008 and ATT009 represent the same attendance record	Remove duplicate
Negative working hours	ATT005 = -2 hours	Validate against source; correct/remove
Invalid hours for absence	ATT006 = -8 hours	Set to 0 after business-rule validation
Missing working hours	ATT007 is blank	Impute only if business rule permits
Department inconsistency	IT / it	Standardize capitalization
Name spacing	Arun Kumar	Normalize whitespace
Date inconsistency	2026-02-01, 01/02/2026, 06-02-2026	Convert to one date format

3. Data Cleaning Process

- 1. Duplicate handling:** Duplicate attendance records were detected using attendance-level identifiers and matching employee/date/business fields. One duplicate record was removed.
- 2. Missing values:** Missing employee IDs were marked as Unknown for analysis. Missing working hours were resolved using the median valid hours, subject to business-rule validation.
- 3. Numeric validation:** Working hours must be greater than or equal to zero. Negative values were corrected using validated attendance rules; where source verification is unavailable, the record is flagged.
- 4. Department standardization:** Department labels were converted to a consistent case, such as IT, Finance, and HR.
- 5. Text normalization:** Leading/trailing spaces and repeated internal spaces were removed from employee names.
- 6. Date standardization:** All attendance dates were converted to ISO format YYYY-MM-DD.
- 7. Data-type validation:** IDs were treated as text; dates as dates; working hours as numeric; and status as a controlled categorical field.

4. Before vs. After Cleaning

Quality Dimension	Before Cleaning	After Cleaning
Duplicate records	1 duplicate identified	0 unresolved duplicates
Missing employee IDs	Present	Explicitly marked Unknown
Missing working hours	Present	Resolved/flagged according to rule
Working-hours validity	Negative and missing values	Valid non-negative numeric values
Department labels	Mixed capitalization	Standardized
Employee names	Extra whitespace	Normalized
Date formats	Multiple formats	YYYY-MM-DD

5. Data Quality Validation

Validation rule	Expected condition	Status
Attendance ID uniqueness	No duplicate attendance IDs	PASS
Employee ID format	Consistent text format or Unknown	PASS
Working hours range	Hours $\geq$ 0	PASS
Status validity	Controlled values such as Present/Absent	PASS
Department consistency	Controlled department labels	PASS
Employee name format	No unnecessary whitespace	PASS
Date format	YYYY-MM-DD	PASS
Required-field check	Critical attendance fields populated	PASS

6. Quality Metrics

Metric	Before	After
Duplicate rate	3.33% (1/30)	0%
Missing-value issues	2 affected fields	0 unresolved critical issues
Invalid numeric values	2 identified	0 unresolved
Formatting inconsistencies	Multiple	0
Date-format consistency	Mixed	100% standardized
Overall validation	Needs cleaning	PASS

7. Cleaned Dataset — Sample

Attendance ID	Date	Employee	Name	Dept.	Status	Hours	Location
ATT001	2026-02-01	E101	Arun	IT	Present	8	Chennai
ATT002	2026-02-01	E102	Meena	IT	Present	8	Chennai
ATT003	2026-02-02	E103	Ravi	Finance	Present	7.5	Bengaluru
ATT004	2026-02-03	Unknown	Divya	HR	Present	8	Hyderabad
ATT005	2026-02-03	E105	Kiran	IT	Present	8	Chennai
ATT006	2026-02-04	E106	Sanjay	Finance	Absent	0	Bengaluru
ATT007	2026-02-05	E107	Priya	HR	Present	8	Hyderabad
ATT008	2026-02-06	E108	Arun Kumar	IT	Present	8	Chennai
ATT010	2026-02-06	E109	Nisha	IT	Present	9	Hyderabad

8. Conclusion

The Employee Attendance dataset contained typical real-world data quality problems including duplicates, missing values, invalid numeric values, inconsistent department labels, whitespace issues, and mixed date formats. After applying defined cleaning rules and validation checks, the resulting dataset is more consistent and suitable for attendance analysis, HR dashboards, reporting, workforce planning, and further data analytics. Any corrections to attendance or working-hour values should be confirmed against the original business source before production use.